Prajwal Mhase IT-B 93

In [49]:
```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from imblearn.over_sampling import SMOTE

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, BatchNormalization
from tensorflow.keras.optimizers import Adam, SGD, RMSprop
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.regularizers import l2
```

In [50]:
```
df = pd.read_csv('Dry_Bean_Dataset.csv').dropna().drop_duplicates()
df['Class'] = LabelEncoder().fit_transform(df['Class'])

X = df.drop('Class', axis=1)
y = df['Class']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

X_train, y_train = SMOTE(random_state=42).fit_resample(X_train, y_train)
```

In [51]:
```
def create_model(optimizer):
    model = Sequential()
    model.add(Dense(64, activation='relu', kernel_regularizer=l2(0.001),
                    input_shape=(X_train.shape[1],)))
    model.add(BatchNormalization())
    model.add(Dropout(0.30))
    model.add(Dense(32, activation='relu', kernel_regularizer=l2(0.001)))
    model.add(BatchNormalization())
    model.add(Dropout(0.30))
    model.add(Dense(16, activation='relu'))
    model.add(Dropout(0.20))
    model.add(Dense(7, activation='softmax'))

    model.compile(optimizer=optimizer,
                  loss='sparse_categorical_crossentropy',
                  metrics=['accuracy'])
    return model
```

In [52]:
```
optimizers = {
    'Adam': Adam(0.001),
    'SGD': SGD(0.01),
    'RMSprop': RMSprop(0.001)
}

early = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
summary = []

for name, optimizer in optimizers.items():
    model = create_model(optimizer)
    model.fit(X_train, y_train, epochs=25, batch_size=256,
              validation_split=0.2, callbacks=[early])

    y_pred = np.argmax(model.predict(X_test), axis=1)
    summary.append([
        name,
        accuracy_score(y_test, y_pred),
        precision_score(y_test, y_pred, average='weighted'),
        recall_score(y_test, y_pred, average='weighted'),
        f1_score(y_test, y_pred, average='weighted')
    ])
```

```
d:\Advanced Data Science Assignments\.venv\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarn
ing: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer u
sing an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Epoch 1/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 13ms/step - accuracy: 0.4794 - loss: 1.5009 - val_accuracy: 0.6752 - val_lo
ss: 1.3750
Epoch 2/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7601 - loss: 0.8345 - val_accuracy: 0.8829 - val_los
s: 0.8763
Epoch 3/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8375 - loss: 0.5989 - val_accuracy: 0.9169 - val_los
s: 0.5364
Epoch 4/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8681 - loss: 0.4927 - val_accuracy: 0.9217 - val_los
s: 0.3829
Epoch 5/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8794 - loss: 0.4433 - val_accuracy: 0.9343 - val_los
s: 0.3036
Epoch 6/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8862 - loss: 0.4150 - val_accuracy: 0.9396 - val_los
s: 0.2720
Epoch 7/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8983 - loss: 0.3830 - val_accuracy: 0.9345 - val_los
s: 0.2756
Epoch 8/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9031 - loss: 0.3687 - val_accuracy: 0.9335 - val_los
s: 0.2736
Epoch 9/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9087 - loss: 0.3486 - val_accuracy: 0.9423 - val_los
s: 0.2476
Epoch 10/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9110 - loss: 0.3427 - val_accuracy: 0.9413 - val_los
s: 0.2495
Epoch 11/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9118 - loss: 0.3317 - val_accuracy: 0.9396 - val_los
s: 0.2474
Epoch 12/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9153 - loss: 0.3210 - val_accuracy: 0.9373 - val_los
s: 0.2508
Epoch 13/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9160 - loss: 0.3131 - val_accuracy: 0.9343 - val_los
s: 0.2520
Epoch 14/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9143 - loss: 0.3123 - val_accuracy: 0.9386 - val_los
s: 0.2401
Epoch 15/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9143 - loss: 0.3100 - val_accuracy: 0.9406 - val_los
s: 0.2245
Epoch 16/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9188 - loss: 0.3002 - val_accuracy: 0.9308 - val_los
s: 0.2540
Epoch 17/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9215 - loss: 0.2950 - val_accuracy: 0.9421 - val_los
s: 0.2281
Epoch 18/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9205 - loss: 0.2906 - val_accuracy: 0.9398 - val_los
s: 0.2235
Epoch 19/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9224 - loss: 0.2883 - val_accuracy: 0.9328 - val_los
s: 0.2488
Epoch 20/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9247 - loss: 0.2786 - val_accuracy: 0.9403 - val_los
s: 0.2219
Epoch 21/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9244 - loss: 0.2798 - val_accuracy: 0.9446 - val_los
s: 0.2078
Epoch 22/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9266 - loss: 0.2757 - val_accuracy: 0.9381 - val_los
s: 0.2228
Epoch 23/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9233 - loss: 0.2813 - val_accuracy: 0.9423 - val_los
s: 0.2085
Epoch 24/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9246 - loss: 0.2742 - val_accuracy: 0.9484 - val_los
s: 0.1942
Epoch 25/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9281 - loss: 0.2655 - val_accuracy: 0.9421 - val_los
s: 0.2087
```

```
85/85 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
Epoch 1/25
```

```
d:\Advanced Data Science Assignments\.venv\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
63/63 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - accuracy: 0.3969 - loss: 1.6120 - val_accuracy: 0.4559 - val_los
s: 1.6904
Epoch 2/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6160 - loss: 1.1368 - val_accuracy: 0.4157 - val_los
s: 1.4352
Epoch 3/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6930 - loss: 0.9687 - val_accuracy: 0.5461 - val_los
s: 1.1814
Epoch 4/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7346 - loss: 0.8592 - val_accuracy: 0.6843 - val_los
s: 0.9519
Epoch 5/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7578 - loss: 0.7842 - val_accuracy: 0.8026 - val_los
s: 0.7571
Epoch 6/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7786 - loss: 0.7255 - val_accuracy: 0.8570 - val_los
s: 0.6177
Epoch 7/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7955 - loss: 0.6760 - val_accuracy: 0.8774 - val_los
s: 0.5367
Epoch 8/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8113 - loss: 0.6360 - val_accuracy: 0.8993 - val_los
s: 0.4720
Epoch 9/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8215 - loss: 0.6018 - val_accuracy: 0.9104 - val_los
s: 0.4239
Epoch 10/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8276 - loss: 0.5813 - val_accuracy: 0.9169 - val_los
s: 0.3882
Epoch 11/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8329 - loss: 0.5619 - val_accuracy: 0.9172 - val_los
s: 0.3702
Epoch 12/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8397 - loss: 0.5414 - val_accuracy: 0.9159 - val_los
s: 0.3595
Epoch 13/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8436 - loss: 0.5221 - val_accuracy: 0.9215 - val_los
s: 0.3422
Epoch 14/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8522 - loss: 0.5084 - val_accuracy: 0.9245 - val_los
s: 0.3200
Epoch 15/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8545 - loss: 0.4917 - val_accuracy: 0.9230 - val_los
s: 0.3205
Epoch 16/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8580 - loss: 0.4834 - val_accuracy: 0.9270 - val_los
s: 0.3141
Epoch 17/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8642 - loss: 0.4713 - val_accuracy: 0.9237 - val_los
s: 0.3113
Epoch 18/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8681 - loss: 0.4598 - val_accuracy: 0.9222 - val_los
s: 0.3137
Epoch 19/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8679 - loss: 0.4551 - val_accuracy: 0.9290 - val_los
s: 0.2945
Epoch 20/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8704 - loss: 0.4469 - val_accuracy: 0.9285 - val_los
s: 0.2958
Epoch 21/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8744 - loss: 0.4396 - val_accuracy: 0.9230 - val_los
s: 0.3008
Epoch 22/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8721 - loss: 0.4389 - val_accuracy: 0.9315 - val_los
s: 0.2785
Epoch 23/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8737 - loss: 0.4387 - val_accuracy: 0.9280 - val_los
s: 0.2920
Epoch 24/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8752 - loss: 0.4235 - val_accuracy: 0.9285 - val_los
s: 0.2831
Epoch 25/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8793 - loss: 0.4230 - val_accuracy: 0.9323 - val_los
s: 0.2756
```

```
85/85 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
Epoch 1/25
d:\Advanced Data Science Assignments\.venv\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
63/63 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - accuracy: 0.4815 - loss: 1.4440 - val_accuracy: 0.7875 - val_loss: 1.3556
Epoch 2/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7837 - loss: 0.7289 - val_accuracy: 0.8900 - val_loss: 0.6897
Epoch 3/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8544 - loss: 0.5076 - val_accuracy: 0.9242 - val_loss: 0.3821
Epoch 4/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8812 - loss: 0.4232 - val_accuracy: 0.9300 - val_loss: 0.3130
Epoch 5/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8938 - loss: 0.3818 - val_accuracy: 0.9353 - val_loss: 0.2767
Epoch 6/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8992 - loss: 0.3529 - val_accuracy: 0.9361 - val_loss: 0.2608
Epoch 7/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9063 - loss: 0.3413 - val_accuracy: 0.9247 - val_loss: 0.2888
Epoch 8/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9115 - loss: 0.3195 - val_accuracy: 0.9343 - val_loss: 0.2543
Epoch 9/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9145 - loss: 0.3103 - val_accuracy: 0.9479 - val_loss: 0.2123
Epoch 10/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9163 - loss: 0.2968 - val_accuracy: 0.9315 - val_loss: 0.2654
Epoch 11/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9155 - loss: 0.3012 - val_accuracy: 0.9434 - val_loss: 0.2167
Epoch 12/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9199 - loss: 0.2952 - val_accuracy: 0.9408 - val_loss: 0.2327
Epoch 13/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9211 - loss: 0.2848 - val_accuracy: 0.9423 - val_loss: 0.2149
Epoch 14/25
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9213 - loss: 0.2852 - val_accuracy: 0.9398 - val_loss: 0.2228
85/85 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
```

In [53]:
```
result = pd.DataFrame(summary, columns=[
    'Optimizer', 'Accuracy', 'Precision', 'Recall', 'F1 Score'
])
result
```

Out[53]:

	Optimizer	Accuracy	Precision	Recall	F1 Score
0	Adam	0.922481	0.922693	0.922481	0.922505
1	SGD	0.911776	0.912080	0.911776	0.911714
2	RMSprop	0.915098	0.917491	0.915098	0.915531